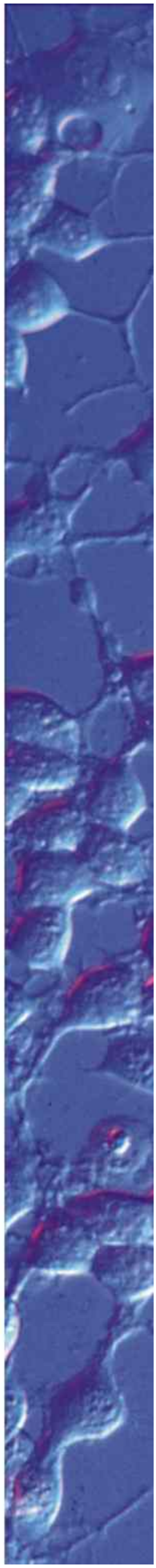
MANUAL
Incubator P 2000



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PeCon Products
Made in Germany

Incubator P 2000

 # 160-800094

Contents

Description

- The INCUBATOR P 2000 is designed for the stabilization of *in vitro* conditions for cell- and tissue cultures during microscopic examination.
- The incubator enables a homogenous heat, CO_2 and O_2 distribution in combination with a Cooling/Heating Insert.
- The heated glass insert warms up the incubation chamber from above. This avoids the condensation of water on the cover of the cell cultivation vessel.
- The heated glass insert of the incubator is permeable to 90% of light in the visible wavelength range.
- The INCUBATOR P 2000 is also suitable for high-resolution microscopy. It is designed for the LD-condensors 0.35 and 0.55.
- For electrical power supply and the control of the temperature inside the incubator, the TEMPCONTROLLER 2000-2 has to be used.
- For the control of the CO_2 concentration, the CO_2-CONTROLLER 2000 has to be used.
- For the control of the O_2 concentration, the CO_2-O_2-CONTROLLER 2000 has to be used.

Safety instructions and hazard warnings

 = *Advice for a trouble-free operation of the device and to avoid its damage.*

 = *Further information for systems optimization.*

- Due to the following conditions, the glass can be hot:
  - during the heat-up phase
  - an ambient temperature below ~ 19°C
  - during cold air motion above the incubator
- Do not feed the incubator with combustible gases, vapors or liquids.
- Do not autoclave the incubator.
- If the incubator is used with local gassing solutions, a sufficient room ventilation has to be ensured to prevent the accumulation of a harmful CO_2-concentration (starting from approx. 0.5 Vol-%) or the accumulation of a harmful N_2-concentration (reducing the oxygen concentration).
- If you have reasons to assume that a safe operation is no longer possible, immediately take the device out of operation and secure it against unintentional use. Reasons to assume that safe operation is no longer possible include:
  - the device shows signs of visible damage,
  - the device does not work anymore,
  - the device has been stored for long periods under unfavourable conditions, or
  - has been subjected to considerable stress in transit.
- You should also follow the additional safety instructions in each chapter of this manual as well as in the manuals of the connected devices.

Usage in accordance with intended purpose

- The product is designed for laboratory use by qualified personnel only.

  It is no medical product according to the IVD Directive 98/79/EG or the MD Directive 93/42/EWG. Therefore, this product is not allowed to be used for medical analysis, diagnostic or treatment purposes on humans within the scope of the IVD or MD Directive.
- The INCUBATOR P 2000 offers a homogenously heated incubation room with a defined atmosphere in combination with following inserts:

PeCon Article-No.	Insert
140-800040	Cooling/Heating Insert P
141-800171	Cooling/Heating Insert P Lab-Tek™

- For power supply and the control of temperature, the CO_2 concentration and the O_2 concentration the INCUBATOR P 2000 has to be used with following devices:

PeCon Article-No.	Device	Parameter
800-800187	TempController 2000-1	°C (Temperature)
800-800037	TempController 2000-2	°C (Temperature)
810-800426	CO_2-Controller 2000	CO_2 (carbon dioxide)
830-800425	CO_2-O_2-Controller 2000	O_2 (oxygen)

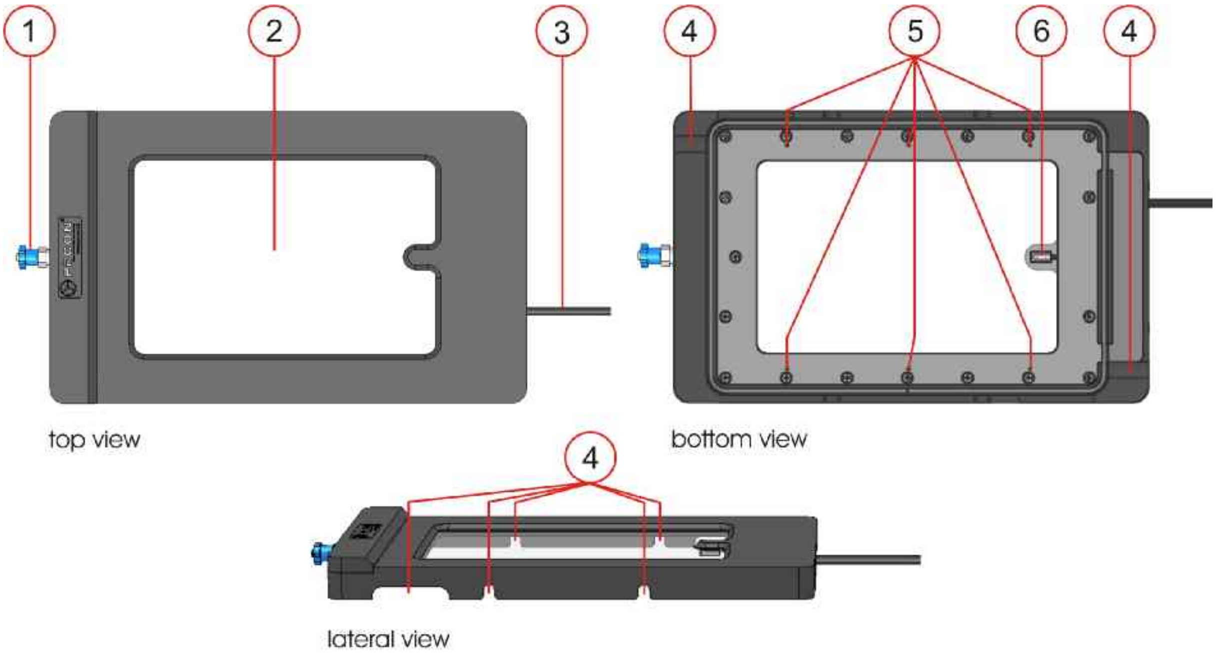
- The heating power is designed for a stable heating up to 30°C above ambient temperature.
- The incubator is designed for increased CO_2 and decreased O_2 values.
- This product is designed for indoor use. Operation is not permitted under unfavourable ambient conditions. The following are unfavourable ambient conditions:
  - dampness or humidity
  - combustible
  - gases, vapours or solvents

- *Use other than the described above will damage the product and other components, used with it. It may additionally involve other risks, such as short circuit, fire and electric shock etc. Do not change or modify any part of the product. The safety instructions must be observed without fail.*
- *Any deviations from the application described here will forfeit all claims against warranty and the manufacturer's liability in general and particularly in the event of a defect. Furthermore, the manufacturer is not liable for consequential damage, unless intention or gross negligence can be proved.*

Scope of delivery

		Qty.	Article-No.
	Incubator P 2000	1	#160-800094
	Operating manual	1	

Operating controls



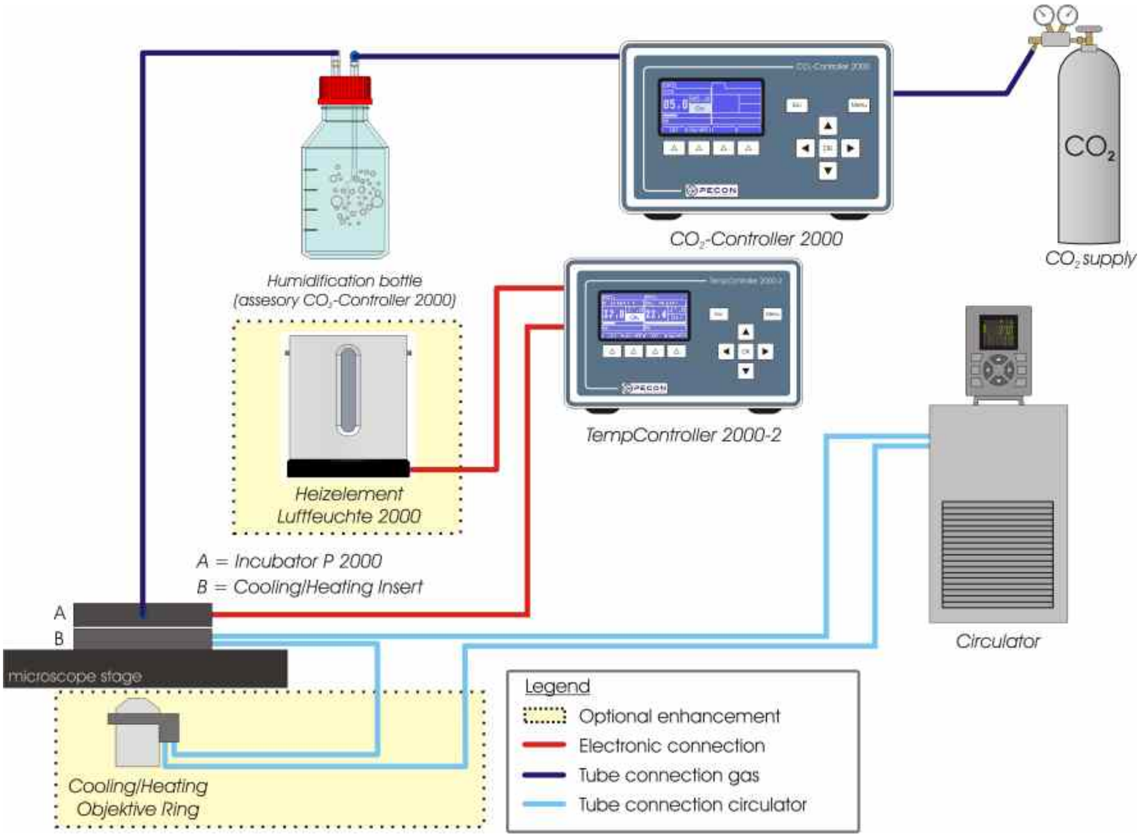
(1) Gas mixture inlet
(2) Heatable glass
(3) Electric connecting cable with 8-pin connector
(4) Openings for cables, perfusion tubes and channel closures of different inserts
(5) Gas mixture outlets
(6) Temperature sensor Pt100

For instructions within this manual, we refer again to the numbers of this legend **(x)***.*

Assembly

Connection diagrams

Application CO_2 control with CO_2-CONTROLLER 2000:



Insertion of the insert into the stage

1. Put the Cooling/Heating Insert (see section "Usage in accordance with intended purpose") into the cut-out (160 x 110 mm) of the scanning stage or mechanical stage.
   Make sure that the spring in the microscope stage clamps the insert.
2. For a constant focus, adjust the insert with the four adjusting screws in the edges of the insert by using the supplied Allen key.

Attaching of the incubator

Put the INCUBATOR P 2000 onto the Cooling/Heating Insert from above.

Connection

1. Connect the Cooling/Heating Insert to the Thermostat (see corresponding manual).
2. Connect the Temperature sensor cable of the INCUBATOR P 2000:
   - Plug the 8-pin connector **(3)** into the channel 2 of the TEMPCONTROLLER 2000-2.

CO_2 control with the CO_2-Controller 2000

The INCUBATOR P 2000 has to be connected to the CO_2-CONTROLLER 2000 and creates the precondition for an optimal pH-value in the culture medium.

> ***Since CO_2 is heavier than the ambient air, it primarily escapes through leaks in the bottom area of the incubation system. Therefore, it is important that the cell cultivation vessels being used cover and seal up the observation opening of the baseplate completely.***

1. Connect the CO_2-CONTROLLER 2000 to the CO_2 supply and the mains supply (see manual CO_2-CONTROLLER 2000).
2. Connect the INCUBATOR P 2000 to the Humidification bottle:
   - Plug the transparent Tygon® tube (supplied with the CO_2-CONTROLLER 2000) on the hose nipple (marked with "incubator") at the Humidification bottle.
   - Attach the other end of the transparent Tygon® tube to the gas mixture-inlet **(1)** of the INCUBATOR P 2000.
3. Connect the Humidification bottle to the CO_2-CONTROLLER 2000:
   - Plug the blue plastic tube (supplied with the CO_2-CONTROLLER 2000) to the blue plug connector of the Humidification bottle.
   - Attach the other end of the blue plastic tube to the plug connector n the rear side of CO_2-CONTROLLER 2000 (see manual CO_2-CONTROLLER 2000).

> *TIP: For a better humidification of the incubation atmosphere, the Humidification bottle can be inserted into the HEATING DEVICE HUMIDITY 2000.*

> *TIP: To avoid the occurrence of a temperature gradient in the cell cultivation medium, mount the COOLING/HEATING OBJECTIVE RING onto the objective.*

Operation

Switch-on

Every heated component has specific properties regarding their heating and cooling behavior. This has to be considered for a stable control of the temperature of this component. To obtain the best adaptation of the loop control to the heated component, the control parameters for channel 1 and channel 2 of the temperature control device have to be configured correctly.

Application with a TempController 2000-1 or 2000-2:

Configure the used channel to "**Device: Inc PM**" with the user interface ("Menu").

For detailed information see the manual of TEMPCONTROLLER 2000-2.

Maintenance

- The INCUBATOR P 2000 is service free, apart from occasionally cleaning.
- The surface of the INCUBATOR P 2000 can be disinfected with a suitable surface disinfectant (e.g. 70% Ethanol).

Do not use any abrasive or chemical detergents or detergents containing solvents.

Do not use cleaners with ethanol. This can lead to stress cracks.

The incubator cannot be autoclaved.

Troubleshooting

By purchasing this device, you have acquired a product that is reliable and operationally safe. However, problems and malfunctions may occur.

Periodically check the technical safety of the device, e.g. check for damage to the housing etc.

Please follow the safety instructions.

The following table shows possible problems, their causes and solutions.

Problem:	Cause / Solution:
1. *The INCUBATOR P 2000 does not heat.*	a) The TEMPCONTROLLER 2000 is not connected to mains power or is not switched on. **Check the setup of the TEMPCONTROLLER 2000 (see manual TEMPCONTROLLER 2000).** b) The main fuse of the TEMPCONTROLLER 2000 is blown. **Check the fuse of the device and replace if necessary (see manual TEMPCONTROLLER 2000).** c) The incubator is not connected to the TEMPCONTROLLER 2000. **Connect the INCUBATOR P 2000 with the TEMPCONTROLLER 2000 (see section "Assembly")** d) The channel(s) of TEMPCONTROLLER 2000 is/are not activated. **Activate both channels of the TempController 2000 (see manual TEMPCONTROLLER 2000).**
2. *The INCUBATOR P 2000 does not reach the nominal temperature.*	a) It takes approx. 10 min to reach a working temperature of 37°C. b) The channel of the TEMPCONTROLLER 2000 is not configured with "Inc PM" (see "Operation"). **Check the configuration on the TEMPCONTROLLER 2000 (see manual TEMPCONTROLLER 2000).**
3. *The cells already die at an early point of time or show unusual behaviour.*	The gas mixture output of the CO_2-CONTROLLER 2000 or the CO_2-O_2-CONTROLLER 2000 is blocked. **Clean it with a higher air pressure (max. 2,5 bar / 36 psi).**

Any other repair must always be carried out by qualified experts familiar with the hazards involved and with the relevant regulations. Unauthorized modifications and repairs of or inside the device will render the warranty null and void.

In case of further questions, consult the technical information service:

e-mail: support@pecon.biz
web: http://www.pecon.biz/pecon/support

Specifications

Material	Acrylic glass (PMMA) black, glass
DC operating voltage	24 V dc protective low voltage
Power consumption	max. 0.5 A
Temperature sensor	Pt100
Control range	Ambient temperature up to 30°C above ambient
Dimensions (WxHxD)	116 x 12.5 x 190 (mm)
Observation area (LxW)	120 x 77 mm
Weight	approx. 0.3 kg